

Descriptive Statistic Project

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Instructor:

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Project:

Descriptive statistics of the average prices of barley and oats in dollars per bushel received by United States farmers for the month of June from 2015 to 2024.

Direct Webpage: <https://www.ers.usda.gov/data-products/feed-grains-database/feed-grains-yearbook-tables#Zipped%20CSV%20files>

Expand: All tables in one file

Click the: Feed Grains Yearbook Data Tables-Recent

Download the Excel file.

Dataset: Value in dollars

Description: Prices of barley and oats in dollars per bushel received by United States farmers for the month of June from 2015 to 2024.

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Table 10--Barley and oats: Average prices received by farmers, United States, dollars per bushel

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	Table 10--Barley and oats: Average prices received by farmers, United States, dollars per bushel																		
2	Commodity and marketing year 1/	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Wt avg 2/					
3	All barley	2015/16	5.04	5.19	5.59	5.49	5.54	5.55	5.88	5.47	5.51	5.43	5.29	5.59	5.52				
4		2016/17	5.39	5.00	4.89	4.54	4.85	4.97	5.12	4.91	5.08	4.97	4.87	5.05	4.96				
5		2017/18	4.46	4.54	4.52	4.32	4.44	4.60	4.38	4.45	4.58	4.46	4.44	4.45	4.47				
6		2018/19	4.62	4.52	4.54	4.49	4.25	4.78	4.64	4.57	4.65	4.88	4.80	4.90	4.62				
7		2019/20	4.82	4.69	4.74	4.56	4.58	4.74	4.51	4.67	4.68	4.89	4.64	4.76	4.69				
8		2020/21	4.57	4.66	4.60	4.76	4.74	4.78	4.79	4.67	4.97	4.90	4.83	4.92	4.75				
9		2021/22	4.86	5.02	5.04	5.11	5.38	5.22	5.41	5.59	5.43	5.69	6.00	6.07	5.31				
10		2022/23	6.44	6.60	7.47	7.51	7.52	7.36	7.48	7.28	7.60	7.50	7.16	7.56	7.40				
11		2023/24	7.39	7.08	7.52	7.51	7.16	7.72	6.75	7.54	7.02	7.54	7.71	7.38	7.39				
12		2024/25	7.04	6.19	6.70	6.73	6.13	6.25	6.45						6.60				

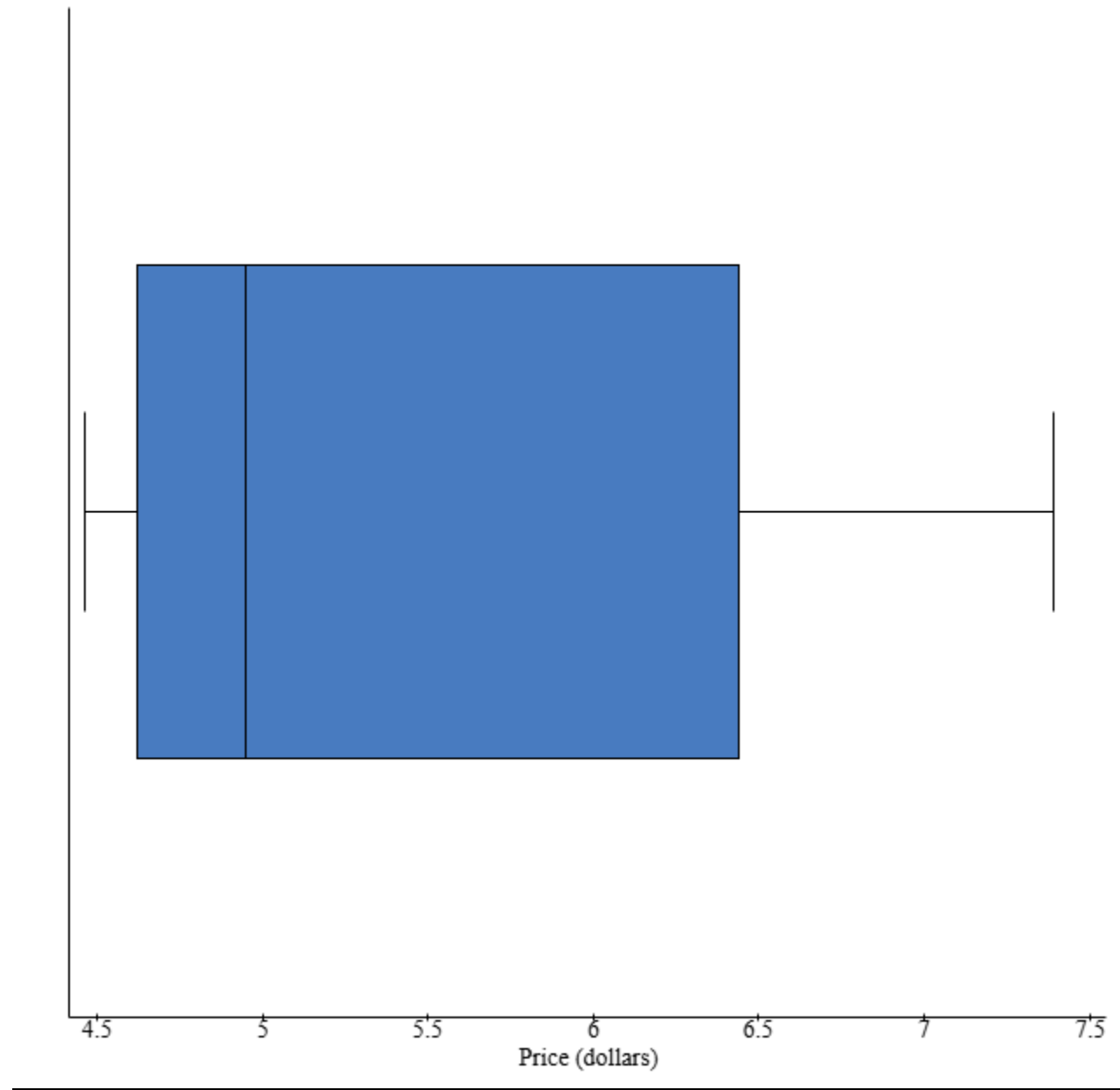
5.04, 5.39, 4.46, 4.62, 4.82, 4.57, 4.86, 6.44, 7.39, 7.04

Objectives:

I shall:

- (1.) Describe the dataset.
- (2.) Present the dataset using histogram.
- (3.) Calculate the measures of center of the dataset.
- (4.) Interpret the measures of center in the context of the dataset.
- (5.) Calculate the measures of spread of the dataset.
- (6.) Interpret the measures of spread in the context of the dataset.
- (7.) Calculate the measures of position (five-number summary) of the dataset.
- (8.) Interpret the measures of position in the context of the dataset.

Data Presentation:



Box-and-whisker plot for the presentation.

Analysis:

- (1.) There are no outliers for the monthly prices of barley and oats per bushel for the months of June 2015 through June 2024.

Let x be the price of barley and oats in dollars per bushel received by United States farmers for the months of June 2015 through June 2024.

Measures of Center:

$$\begin{aligned}\Sigma x &= 5.04 + 5.39 + 4.46 + 4.62 + 4.82 + 4.57 + 4.86 + 6.44 + 7.39 \\ &\quad + 7.04 = 54.63\end{aligned}$$

$$n = 10$$

$$\text{Mean, } \bar{x} = \frac{\Sigma x}{n} = \frac{54.63}{10} = \$5.463$$

Mean: The average price of barley and oats per bushel received by United States farmers for the months of June 2015 through June 2024 is approximately \$5.46.

I am going to sort the data in ascending order.

Dataset in ascending order is:

4.46, 4.57, 4.62, 4.82, 4.86, 5.04, 5.39, 6.44, 7.04, 7.39

$$\text{Median} = Q_2 = \frac{4.86 + 5.04}{2} = \frac{9.9}{2} = \$4.95$$

Median: When the monthly prices of barley and oats per bushel for the months of June 2015 through June 2024 is sorted in ascending order, the median price is \$4.95.

Mode: There is no price value that occurred most frequently in the dataset.

The minimum price is \$4.46

The maximum price is \$7.39

$$\text{Midrange} = \frac{\text{minimum} + \text{maximum}}{2} = \frac{4.46 + 7.39}{2} = \$5.925$$

Midrange: From 2015 to 2024 in the months of June only, the average price of the minimum price and maximum price of barley and oats per bushel is approximately \$5.93 in the United States.

Measure of Spread:

Let us calculate the range

Range = maximum – minimum

$$= 7.39 - 4.46$$

$$= \$2.93$$

Range: The difference between highest price of barley and oats per bushel and minimum price of barley and oats per bushel received by United States farmers for the months of June 2015 through June 2024 is \$2.93.

Next, we calculate the variance of the dataset

x	$x - \bar{x}$	$(x - \bar{x})^2$
5.04	-.423	.178929
5.39	-.073	.005329
4.46	-1.003	1.006009
4.62	-.843	.710649
4.82	-.643	.413449
4.57	-.893	.797449
4.86	-.603	.363609

6.44	.977	.954529
7.39	1.927	3.713329
7.04	1.577	2.486929
$\Sigma(x - \bar{x})^2 = 10.63021$		

$$\text{Variance, } s^2 = \frac{\Sigma(x - \bar{x})^2}{n - 1} = \frac{10.63021}{10 - 1} = 1.181134444\2$

$$\text{Standard Deviation, } s = \sqrt{\text{Variance}} = \sqrt{1.181134444} = \$1.086800094$$

Standard Deviation: This value of the standard deviation tells us the price is not too distant (not spread) from the mean. There is little spread among the price.

Interquartile Range:

Next, we shall calculate the Interquartile Range.

This is the difference between the Upper Quartile and the Lower Quartile.

Dataset in ascending order is:

4.46,4.57,4.62,4.82,4.86,5.04,5.39,6.44,7.04,7.39

$$n = 10$$

$$\frac{1}{4} * 10 = 2.5$$

Then I rounded to the 3rd position, which is the 3rd number in the ascending order

Lower Quartile, $Q_1 = \$4.62$

$$\frac{3}{4} * 10 = 7.5$$

Then I rounded to the 8th position, which is the 8th number in the ascending order

Upper Quartile, $Q_3 = \$6.44$

Interquartile Range, $IQR = Q_3 - Q_1$

$$IQR = 6.44 - 4.62 = \$1.82$$

Interquartile Range: The variation in 50% of the price of barley and oats in dollars per bushel received by United States farmers for the month of June from 2015 to 2024 is \$1.82.

Measures of Position:

The Five-Number summary of the dataset and their interpretations are:

$$\textit{Minimum} = \$4.46$$

The minimum price of barley and oats in dollars per bushel received by United States farmers for the month of June from 2015 to 2024 is \$4.46.

$$\textit{Lower Quartile}, Q_1 = \$4.62$$

25% of the price of barley and oats in dollars per bushel received by United States farmers for the month of June from 2015 to 2024 is below \$4.62.

The 5th position is 4.86

The 6th position is 5.04

$$\frac{4.86 + 5.04}{2} = 4.95$$

$$\textit{Median} = \$4.95$$

50% of the price of barley and oats in dollars per bushel received by United States farmers for the month of June from 2015 to 2024 is \$4.95.

Upper Quartile, Q_3 = \$6.44

75% of the price of barley and oats in dollars per bushel received by United States farmers for the month of June from 2015 to 2024 is below \$6.44.

Maximum = \$7.39

The maximum price of barley and oats in dollars per bushel received by United States farmers for the month of June from 2015 to 2024 is \$7.39.

References Page (MLA)

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